

ARCHITECTURE PORTFOLIO

AHMAD FIQRY FUAD



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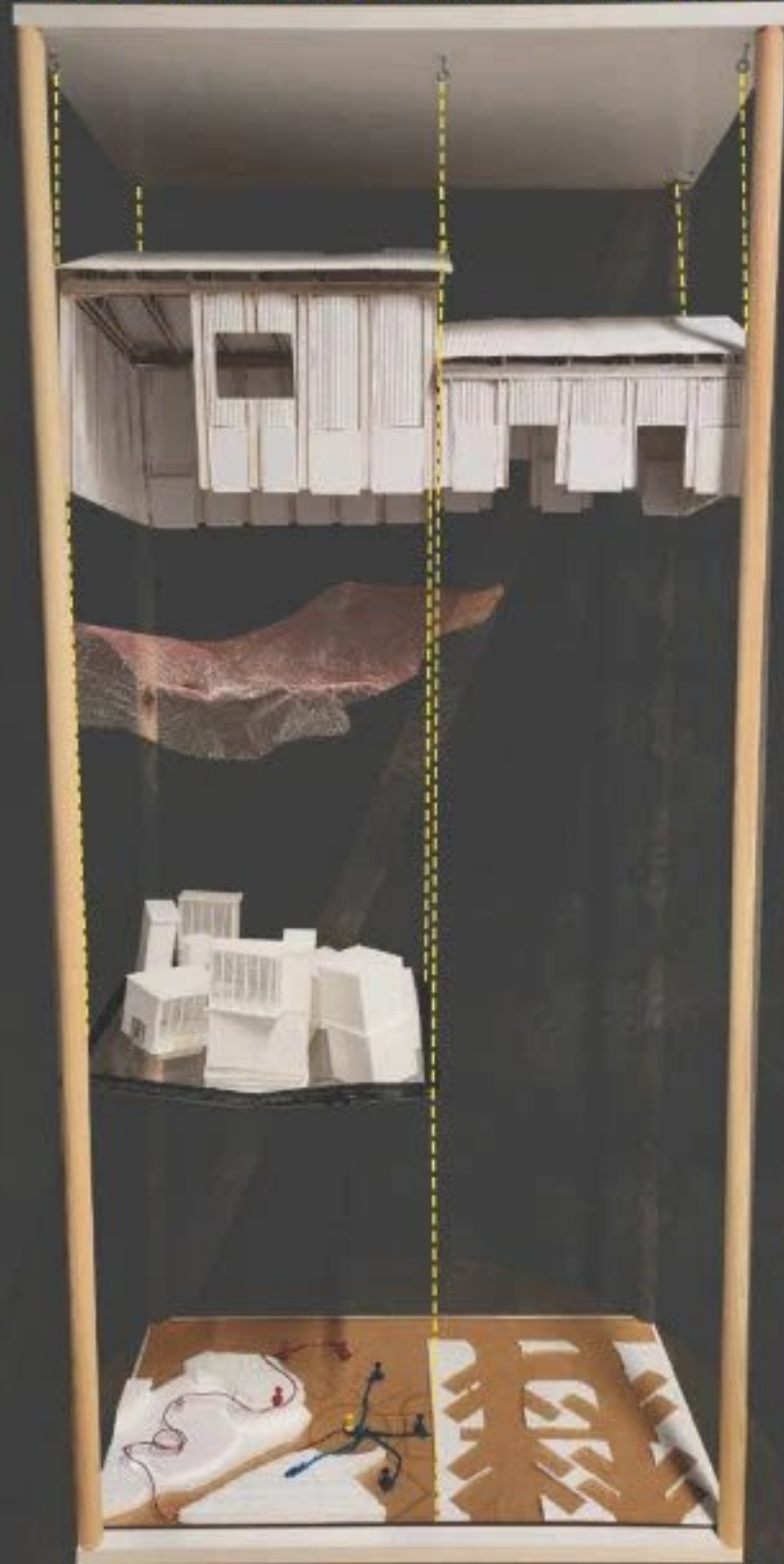
EITRI

HOUSE
OF
METAL

AHMAD FIQRY FUAD



EXPLODED PHYSICAL MODEL



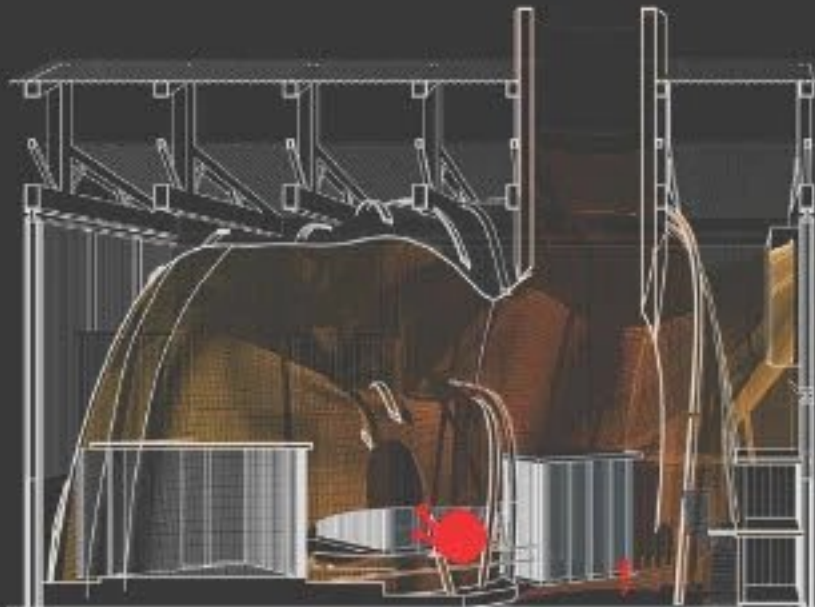
DESIGN CONCEPT

The concept of this project is to create a temporary and interactive between the recycling process of metal and the visitors. The idea is to connect the users with a certain experience. This experience will be a flow of a mechanical part that can be driven and controlled by the user, in the beginning until a certain point.

This created experience would aim to engage the visitor with the activity on the site and give them a hands-on view on a mechanical to bring home.

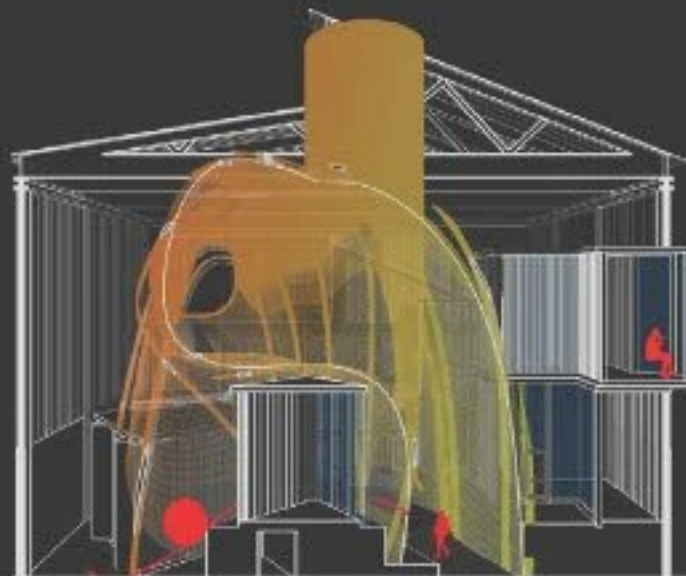
The metallic mesh made with recycled metal is used to guide and control the main place. The mesh also guides the user and run into the site. This comes in contact with the warehouse that covers the mesh. The mesh design is patterned in a way to open up during the winter.



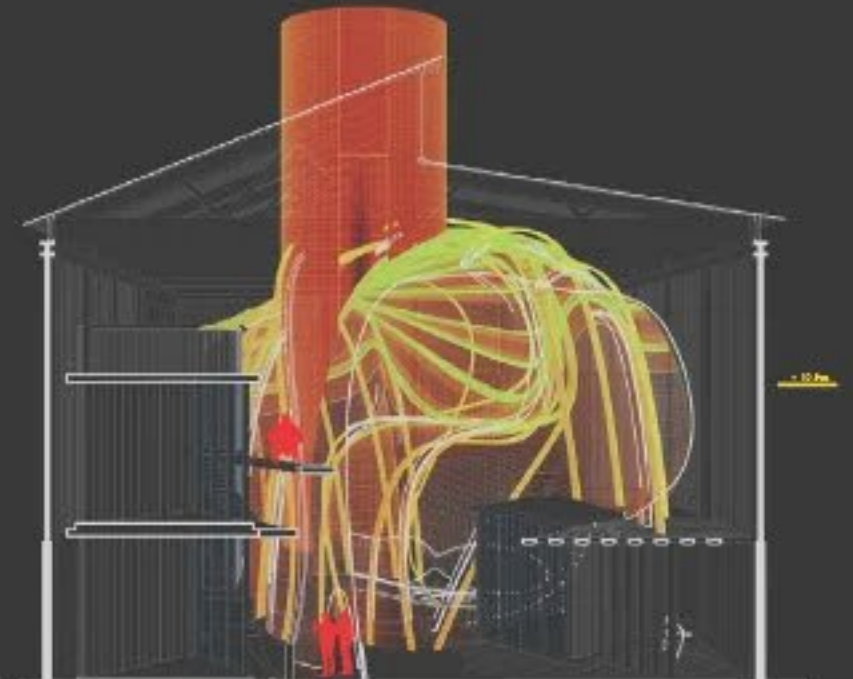


SECTION AT C

ALL SECTIONS FOLLOW THE SAME SCALE BAR



SECTION AT 4

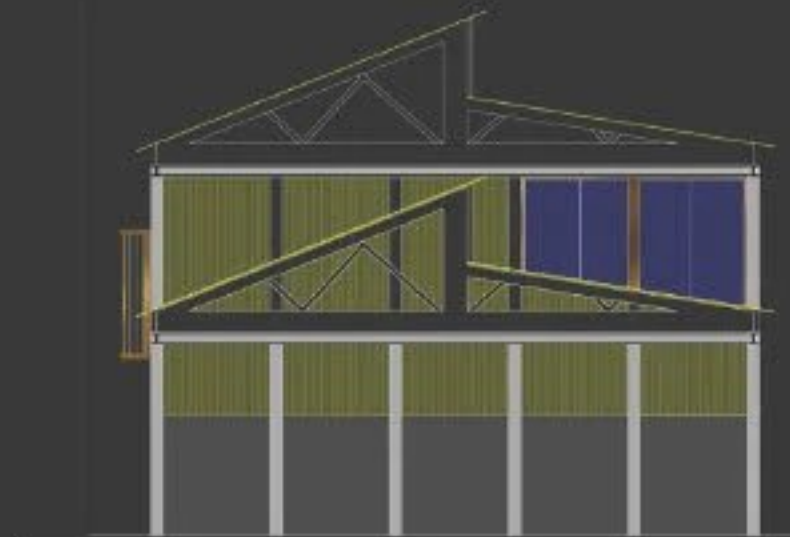


SECTION AT 6

10 m



FRONT ELEVATION



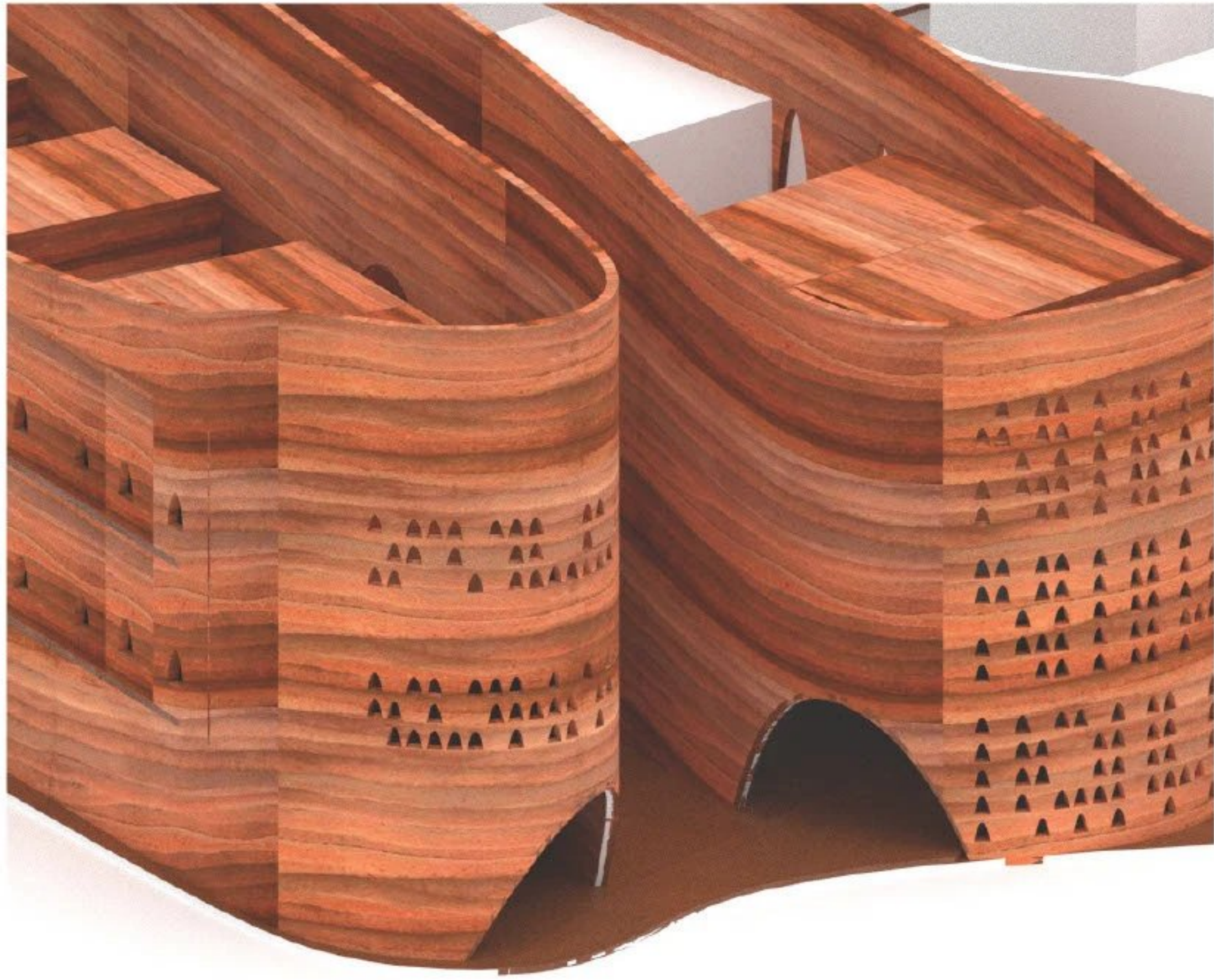
RIGHT ELEVATION





meisa RESIDENCE

AFFORDABLE HOUSING COMPLEX



ABSTRACT WIND STUDY



Wind flow attributes - expansion to compression to release. Morphing of speed and volume causes flow that resembles a tied knot.



Cloth tied by pins, twisted and spread around - study of wind's ability to spread and stretch from a small knot to larger wider spaces.



Wind circulation study - wind circulates into loops and flow similar to a circuit. In the wider scale and flow from one place to another and comes back annually. This study finds the nature of the circulation in a smaller scale.



Al Barsha Social interaction abstract study - The people of Al Barsha are connected through nodes—key zones where interaction occurs. These nodes bring together individuals from diverse backgrounds, fostering connections within the community.



Social interaction abstract study 2 - In Al Barsha, people frequently cross paths but rarely engage with one another, resulting in a subtle yet noticeable segregation.



Site typology abstract study - The site's towering typology looms over its residents, creating a physical barrier that feels engulfing. However, the presence of numerous open plots acts as a gravitational pull, drawing Al Barsha's occupants to these spaces for activities and as a reprieve from the imposing buildings.

VOLUMETRIC SPACIAL STUDY



Al Barsha Social interaction abstract study - The people of Al Barsha are connected through nodes—key zones where interaction occurs. These nodes bring together individuals from diverse backgrounds, fostering connections within the community.



Social interaction abstract study 2 - In Al Barsha, people frequently cross paths but rarely engage with one another, resulting in a subtle yet noticeable segregation.

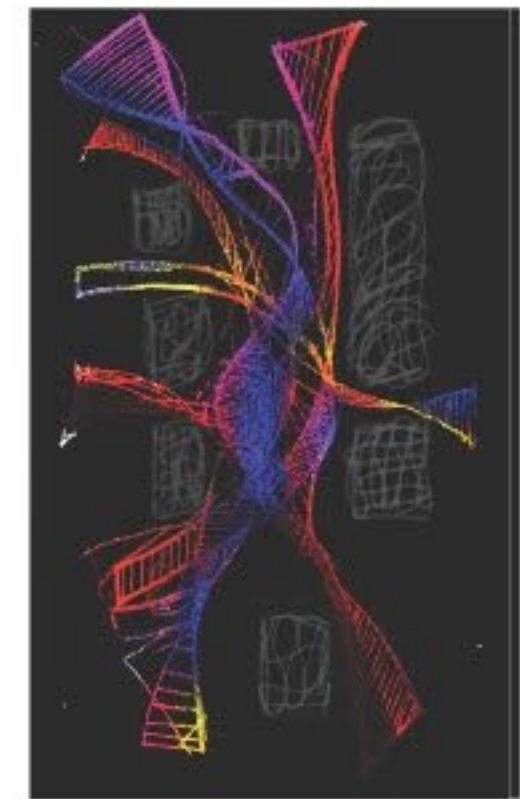


جاشنمال
JASHANMAL

NARRATIVE

Jashanmal Corporate employees often work long office hours with minimal physical activity, leading to potential long-term health issues such as dietary concerns and organ-related conditions. The company's primary age group, between 20 and 45 years old, shows a strong desire to adopt a more active lifestyle. However, the lack of accessible sports facilities and parks in Al Barsha severely limits their ability to do so. The absence of recreational spaces near their homes forces many employees to forgo regular physical activity. This issue is compounded by their demanding work schedules, which leave little time to travel to distant amenities. Additionally, the higher rental costs of properties near parks or sports centers make such options unaffordable for most employees, particularly those in administrative roles. As a result, employees often resort to using the area's vacant lots for recreation. These open spaces, characterised by barren, sandy terrain, are not only unsuitable for physical activity but also pose risks such as exposure to traffic and lack of privacy. This restrictive environment creates a monotonous routine, confining employees to a cycle of commuting between work and home, leaving their need for a balanced and active lifestyle unfulfilled.

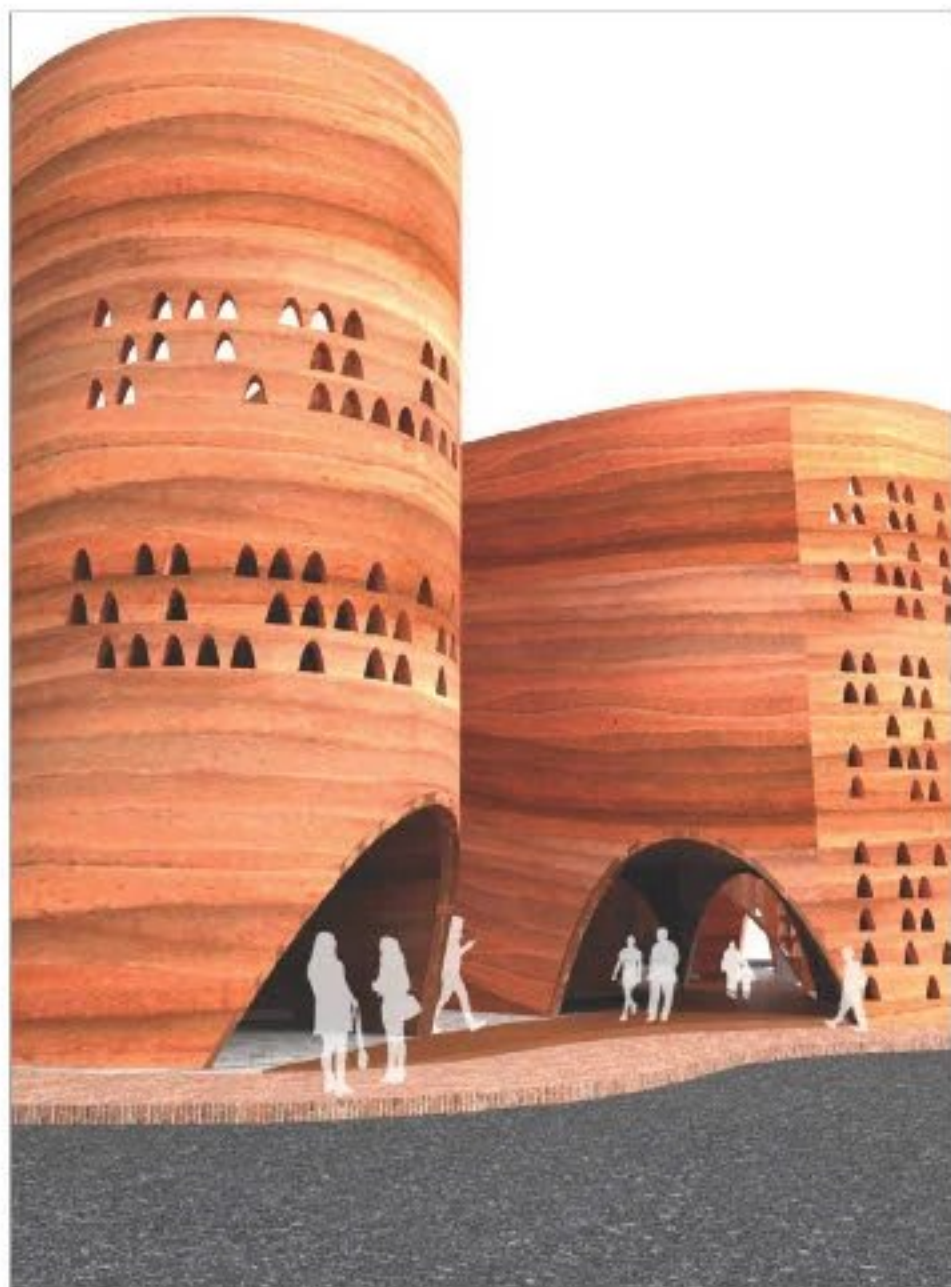
Building design position:
TO CREATE FAMILIAR SPACES THROUGH PHYSICAL ACTIVITIES



ARCHITECTURAL DESIGN
URBAN FORCES
TANGIBLE AND INTANGIBLE, WHAT AND WHY?

SECOND FLOOR

The second floor are the full private zone of the building. The floor comprise of larger units which are one and two bedroom units. on the floor there are four two bedroom units that faces the multi-purpose court. This is done to allow the residence to view the activity at the ground floor despite being the furthest away from the activity. The larger units are placed on the second floor to allow better privacy from noise and crowds. It is more suitable for those who has families up to two children.



2ND FLOOR PLAN



Arched openings held by CCS concrete beams paves a dedicated main straight that the residents of Al Barsha can walk through. As they walk through this walk way the walls would direct their view towards the main activity happening within that row. This intern would create the effect of familiarity for visitors that often walks pass or occupants of the building where they experience this unique view on the daily basis.

ARCHITECTURAL DESIGN FINAL DESIGN

MODEL PICTURES, RENDERS, CONSTRUCTION DETAILS



SITE PLAN

At the site plan level, the building integrates with its surroundings by offering facilities that are accessible to the community. Given the lack of sports amenities and play areas for children in Al Barsha, this building becomes a focal point for residents seeking recreation and relaxation. Some visitors may use the space for leisure, while others may utilize it as a venue to challenge themselves physically. Providing such amenities activates the site throughout the day, including during working hours and midday. Families, children, and mothers can enjoy the facilities for a nominal fee, which contributes to the space's maintenance and utility costs, ultimately reducing the overall rent expenses for residents.

The building's curvilinear form establishes it as a new landmark in Al Barsha, complementing the area's most notable existing landmark, the Mall of the Emirates. This distinctive architectural feature enhances the site's activity levels and attracts tourists visiting the area. Familiarity is fostered through the building's design, which harmonizes with the site's views, occupants, users, and functional offerings.

ARCHITECTURAL DESIGN FINAL DESIGN

MODEL PICTURES, RENDERS, CONSTRUCTION DETAILS

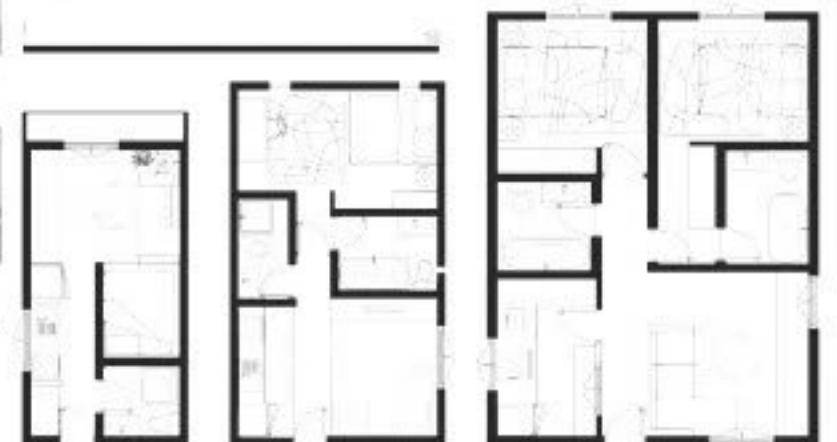


SOUTH ELEVATION
SCALE 1:500



NORTH ELEVATION

UNIT PLANS

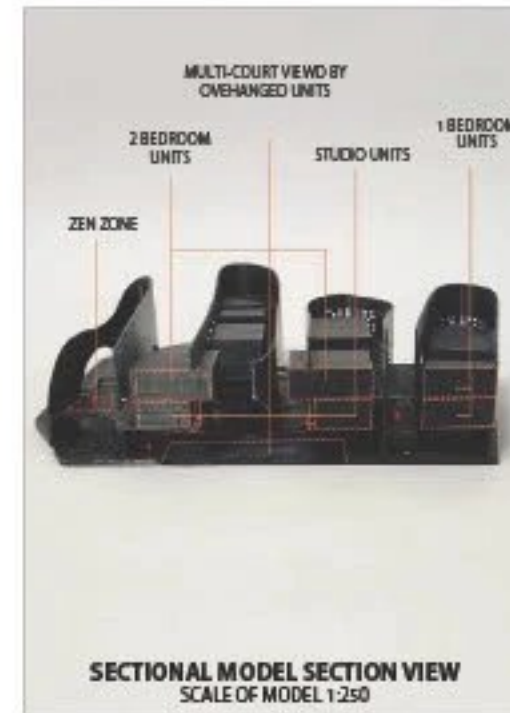
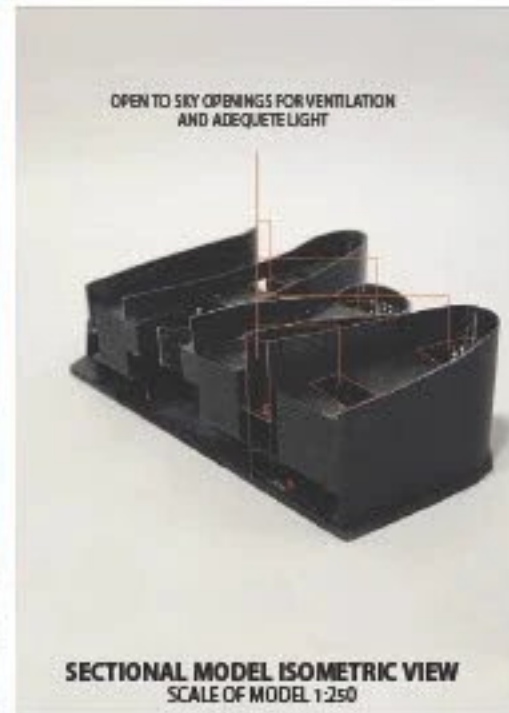




CROSS VENTILATION
DIAGRAM



SECTION AA



SECTION VIEW

The building's spaces are better understood in a section. The corridors coupled with tall walls allows for passive cross ventilation to take place. The spaces between units and sports zones can be viewed by one another as the units are above the sport courts they prevent any privacy leakage. There are several open to sky zones in the building making sure sunlight penetrates the masses and each floors. Observation in section suggest the pace of each floor due to the intended activity by design. The ground floor are fast pace zones catering to movement and flow of the public whereas the zones above are slower and more concise. They follow a different rhythm allowing for a comfortable living condition despite the loud activities.



SECTION BB



SECTIONAL ZONING DIAGRAM

HERIOT-WATT SCHOOL OF ARCHITECTURE

Edinburgh - 2025
Site - Royal Infirmary (pre-renovation)
Original Architect - David Bryce
Site area - 3000sqm
Site's current building - Edinburgh Futures Institute by Bennetts Associates

The Royal Infirmary first built in 1879 by a Scottish architect David Bryce. It is situated on Lauriston Place where neighboring landmarks such as the George Heriot school sits right across the road. The royal infirmary has been demmed the UNESCO World Heritage site where it is graded category A building. The building plays a significant role for Edinburgh as it sits to be one of the longest lasting building in the site.

The Royal Infirmary was abandoned in 2003 as it relocates itself to Little France. This meant that the building was in a poor state before Bennetts as-ociates intervene. The walls stood up straight and fine with cracks in a few places. However the condition of the building's floor was not the same. It suf-fered from large water leaking over the years pooled up to cause warping on the floors and structural beams of the building. The pooled up water covered an entire floor as it slowly erode the tiles away as years goes by.

Bennetts associates took over the building as the city receive funding from the locals. The main challenge the building undertook was renovating the interior heavily as mold, water damage, rott and so many other defects took a toll on the building. Bennetts associates played a crucial role in bringing this building back together as it now stands as an open student hub for all to come and visit owed by the University of Edinburgh. It is situated right beside the Meadows which is a significant open field that the community actively in-teracts with. Bennetts associates uses a completely different approach to the original design concept that David Bryce used. To integrate and open instead of segregate and separte.

The project is to imagine the building prior to Bennetts associates. Using the runned down old building. A new school of Architecture owned by Heriot-Watt University will be open to all. Exploration of using provocative or conservative methods has been done in order to satisfy the design concept.

D69FA - 2025 - EDINBURGH
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TUTORS - MANUEL CRESCIANI (STUDIO LEAD) - ALEX MACLAREN (CO LEAD) - SOFIA LIBERALI (CO TUTOR)

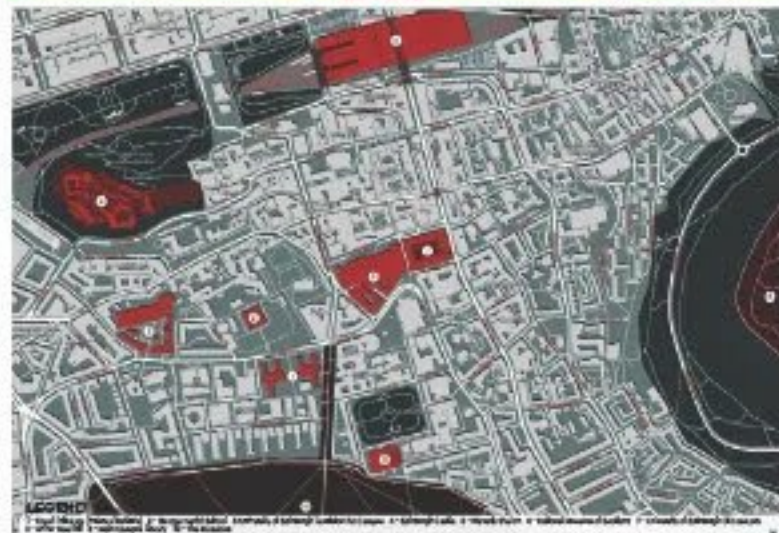
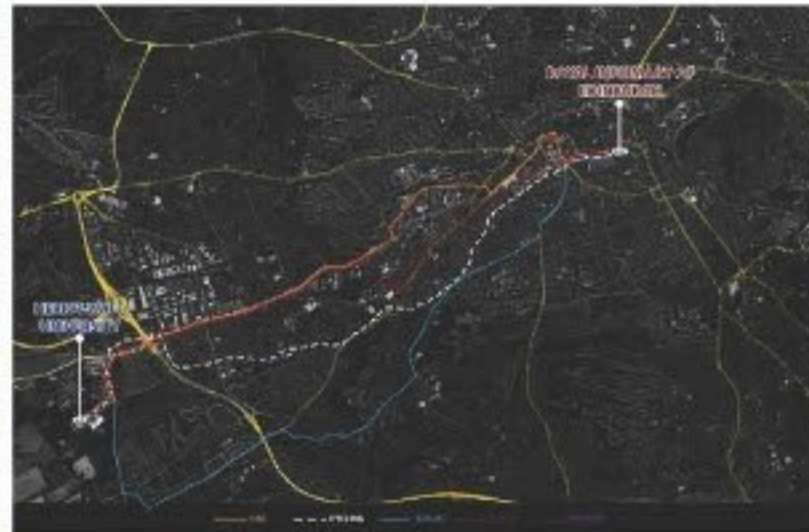
URBAN CONTEXT

THE SITE AND IT'S PERKS

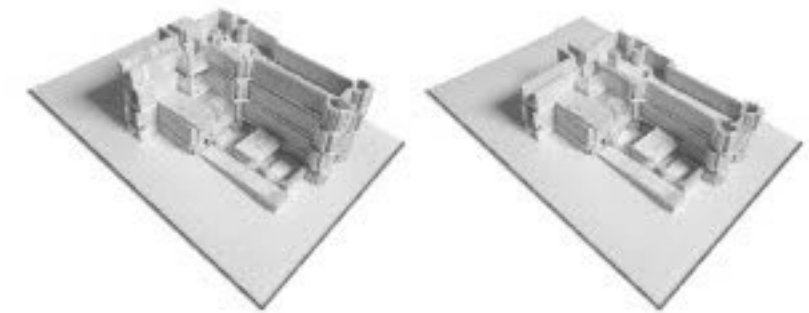
Edinburgh is a walkable city; this is no exception to the site of the Royal Infirmary. The Infirmary site is in the centre where the visitors can walk a maximum of 15min to reach key landmarks around Edinburgh. Additionally, bus systems connects the site to the main parts of the city as well as the current Herit-Watt campus. It is only 30mins away, allowing students to travel for free and easily to the site. The site surrounds itself with shops and cafes resulting the the lack of need for the amenities on site. However, the ability to offer would allow the building to run and earn its bull even in off-hours and low peak times.

The Futures Institute now acts as an urban hub that allows visitors to enjoy a warm indoor space inclusive of all ages. The building's position besides a main bus stop and the Meadows makes it a frequent stop spot for those requiring a break from the walkable city.

At peak hours the students of the George Heriot school swarms the front of the building trying to catch the next bus. This provides ample of opportunities to intrigue the students with the activities happening inside the building resulting to a pubic attraction.

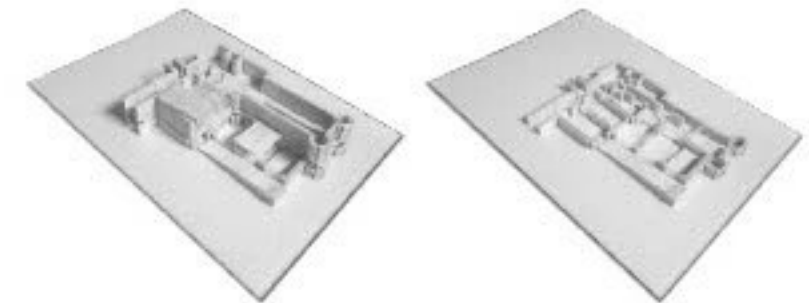


DESIGN ITERATION 3 1:200 MODEL PICTURES



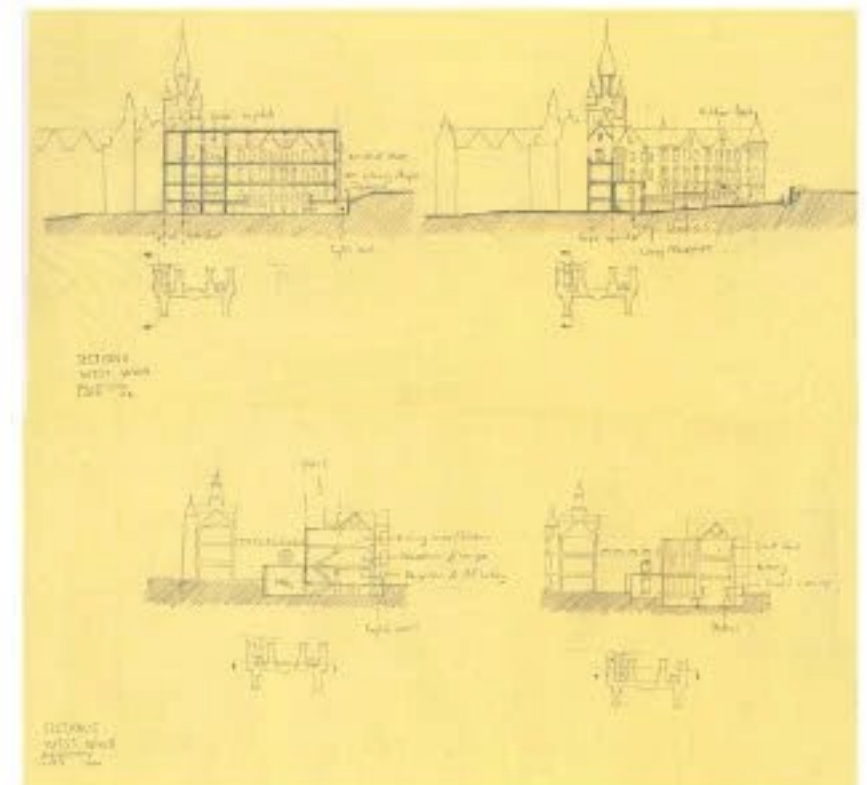
2ND FLOOR

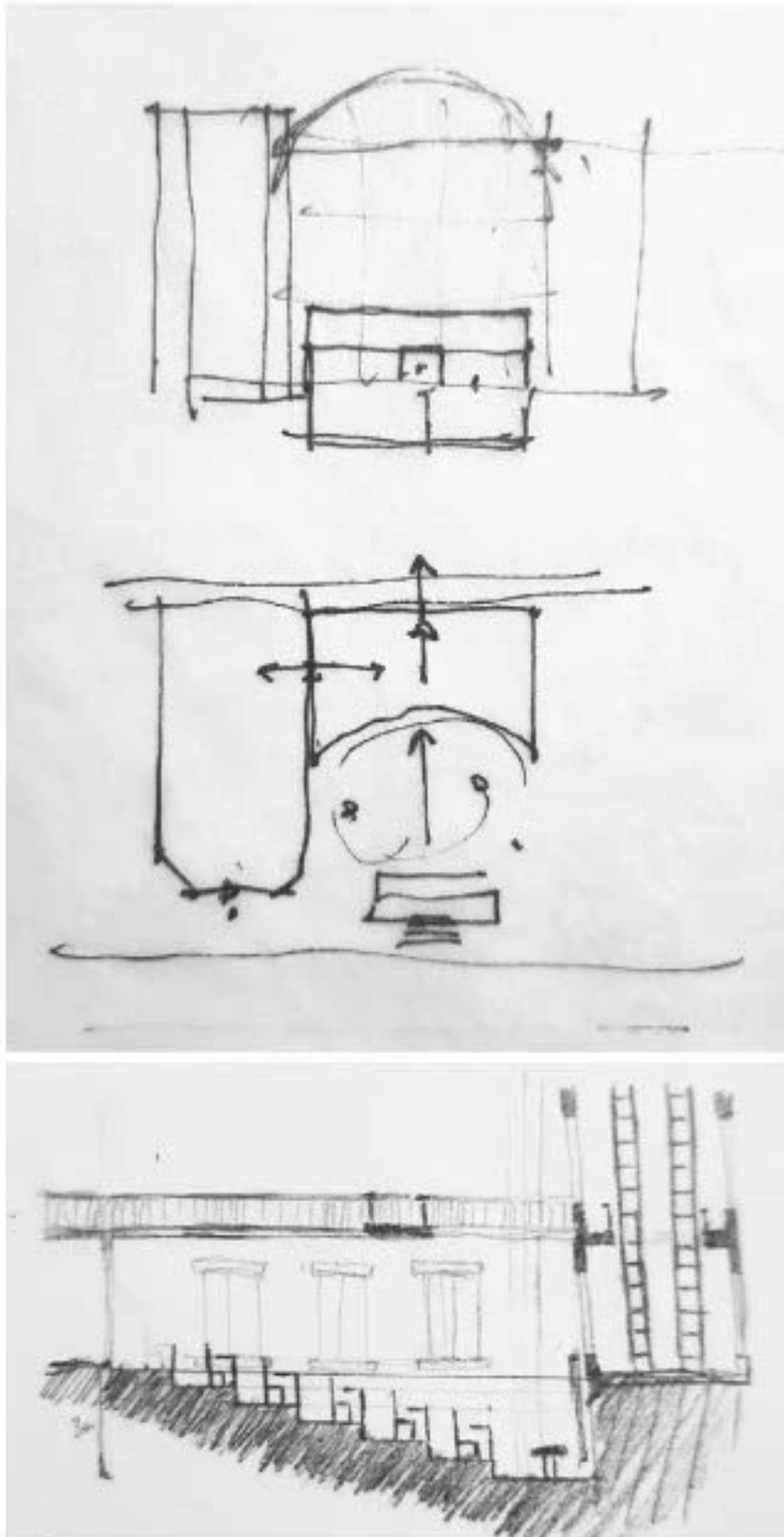
1ST FLOOR



GROUND FLOOR

LOWER GROUND FLOOR





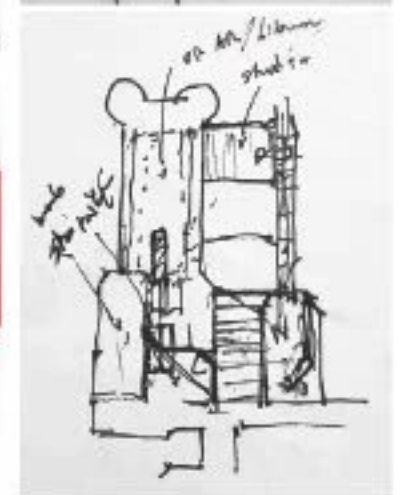
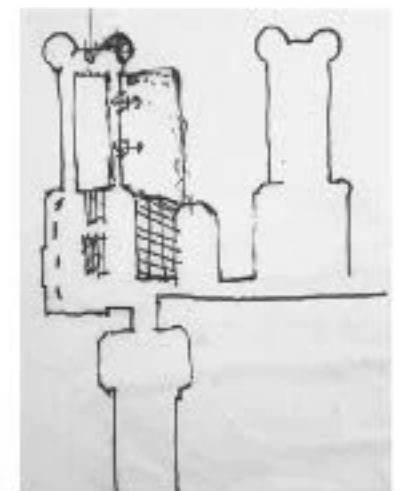
DESIGN ITERATION 3

HONORING THE OLD AND BREAKING NORMS

Design Iteration Two failed due to its significant intervention in the exterior of the building. The design largely concealed the old façade, which conflicted with preservation requirements. UNESCO declared the building a Grade A listed structure, meaning it should be preserved as much as possible. As a result, a new design concept emerged—one that sought to honour the existing façade and make minimal changes to the exterior, while allowing the freedom to completely renovate the interior.

The first development of Iteration Three centred around excavating the ground to expose the basement floor. This approach was part of the ongoing idea to respect and reveal the old structure. Studio spaces were proposed alongside the original basement walls, connected as part of the interior flow.

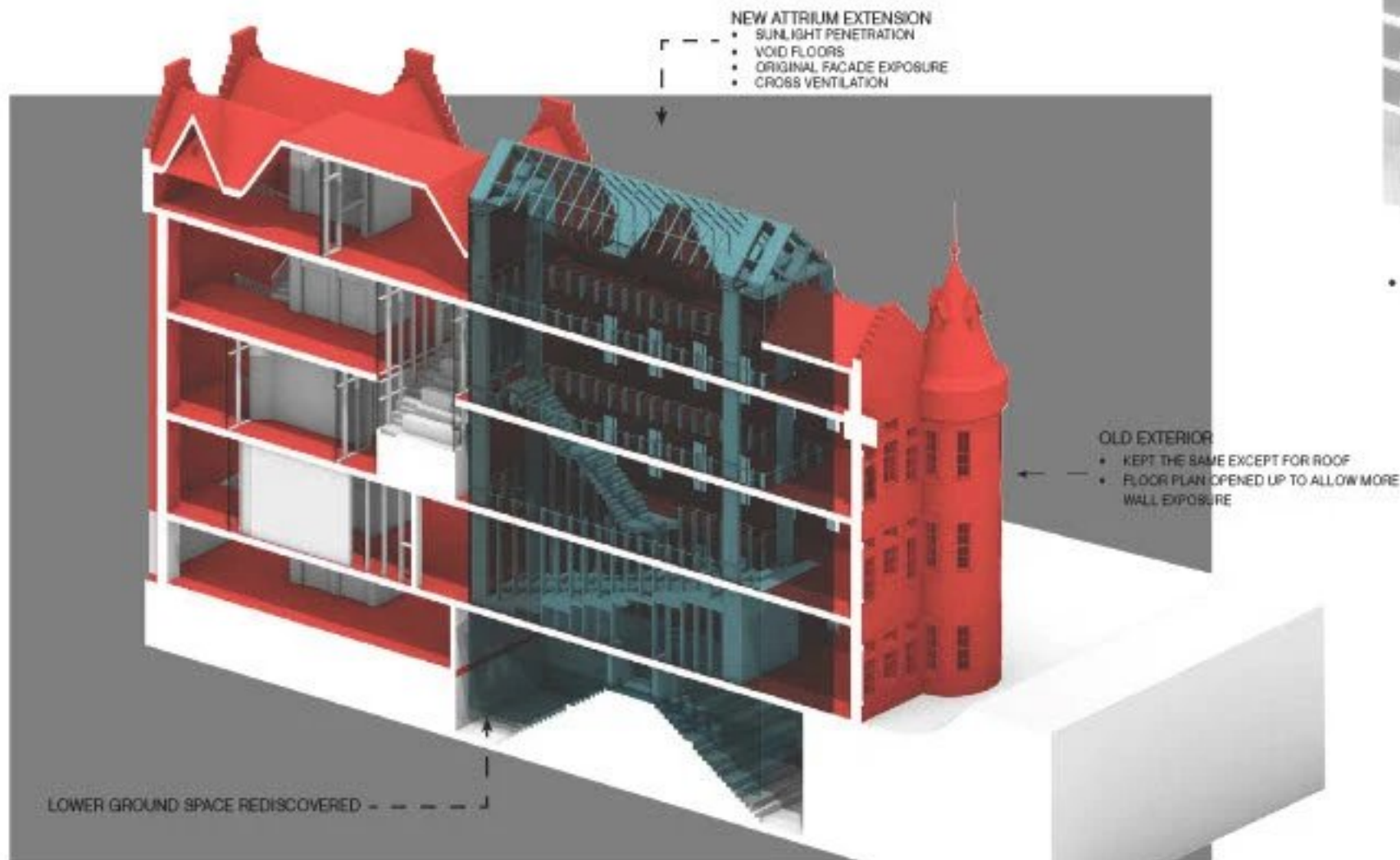
However, this idea was later reconsidered. Excavating such a large portion of land to insert new spaces was ultimately deemed unsustainable and not cost-effective.



• Carved out land repurposed for studio spaces

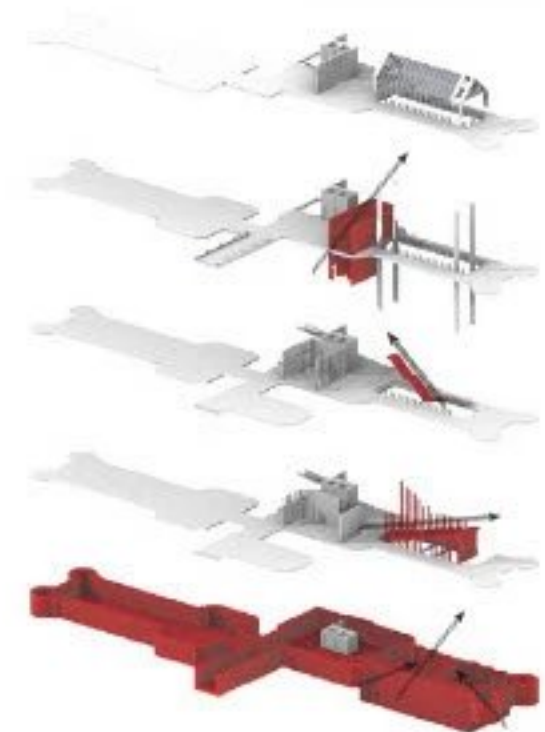
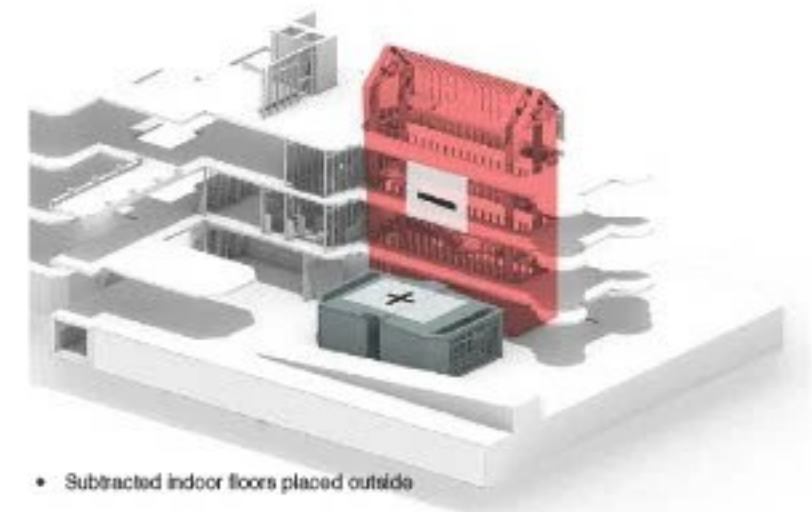
• Current ground removed to open up the basement

FINALISED DESIGN CONCEPT



HONORING THE OLD CONCEPT

Concept to break and connect space three dimensionally whilst keeping the original facade untouched.



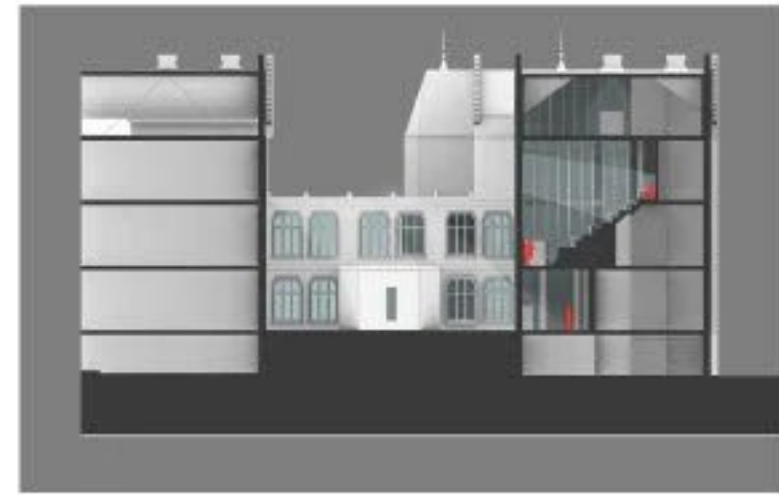
SECTION & SUSTAINABILITY STRATEGY



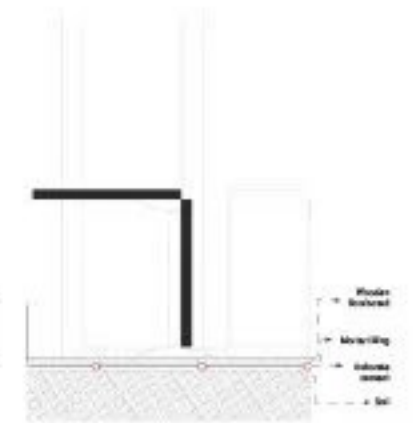
SECTION 2



UNDER FLOOR HEATING PLAN LAYOUT



SECTION 3



UNDER FLOOR HEATING SECTION

SECTION & SUSTAINABILITY STRATEGY

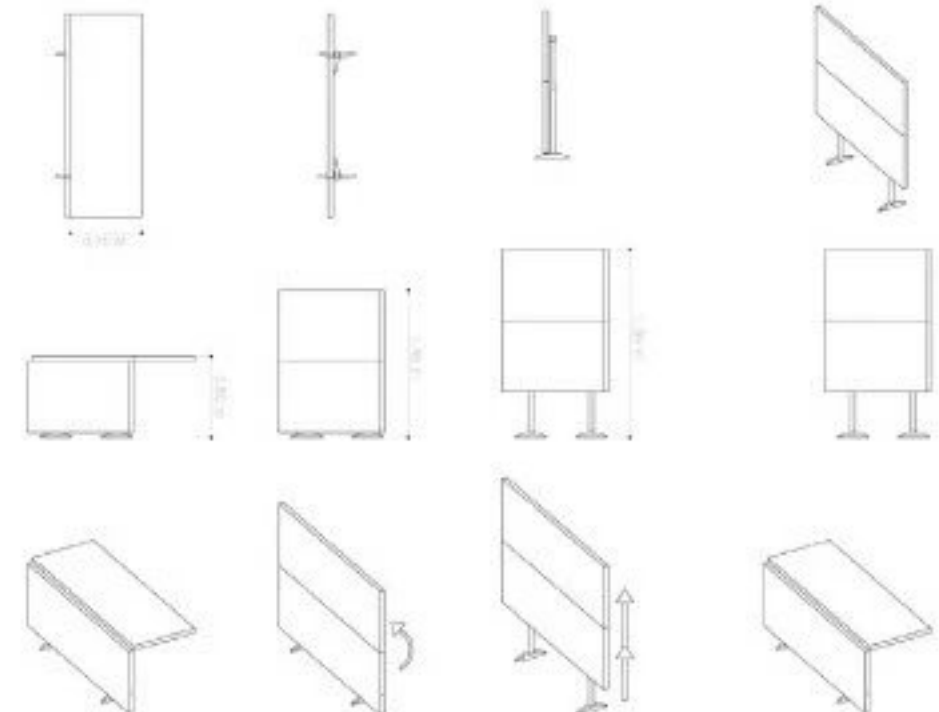


SECTION 1



CROSS VENTILATION STRATEGY

STUDIO FURNITURE



PLANS



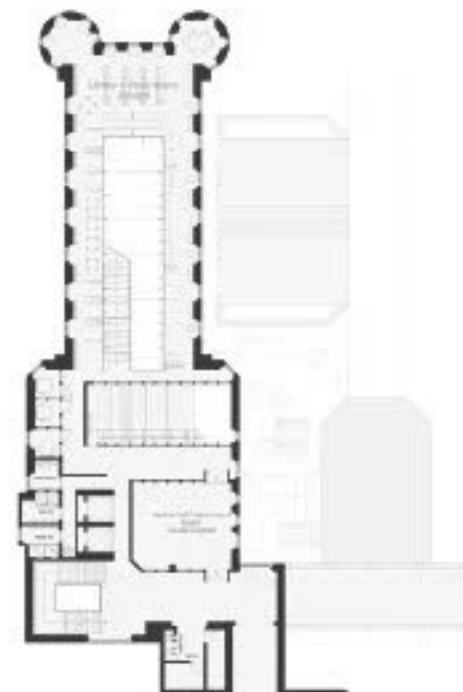
LOWER GROUND FLOOR PLAN



GROUND FLOOR PLAN



1ST FLOOR PLAN



2ND FLOOR PLAN



ATTIC FLOOR PLAN
SCALE 1:200

